

Reproduction Study: Pixel Error-Value Ordering with a CNN Predictor for RDH (Saini & Singh, 2025)

IIIT Vadodara Internship reproduction pipeline

Original work: Ankit Kumar Saini, Samayveer Singh, *Signal, Image and Video Processing*, vol. 19, 2025

Abstract—We reproduce the reversible data hiding principle of Saini and Singh, which combines pixel error-value ordering (PEVO) with a CNN-based predictor. As no trained CNN or its weights are available, we substitute a fixed rhombus predictor as a stand-in learned predictor and reproduce the error-ordering PEE mechanism, its capacity-distortion behaviour, and exact reversibility. The CNN component is therefore approximated, which is stated openly.

Index Terms—reversible data hiding, prediction-error, pixel error value ordering, CNN predictor

I. INTRODUCTION

A better predictor sharpens the prediction-error histogram and raises capacity. Saini and Singh use a CNN predictor with pixel error-value ordering. Lacking the trained network, this report reproduces the PEVO/PEE mechanism with a fixed predictor as a stand-in and reports the resulting trade-off, clearly labelling the CNN as approximated.

II. RELATED WORK

PEE (Thodi-Rodriguez), sorting+prediction (Sachnev), and pixel-value-ordering (PVO, Li et al. 2013) are the lineage. CNN predictors are a recent trend for sharper error histograms.

III. METHODOLOGY

Pixels are predicted (here by the rhombus double-set predictor as a stand-in for the CNN); prediction errors are ordered and the innermost bins expanded to embed. Ordering by error magnitude concentrates embedding in the most predictable pixels. A trained CNN predictor would further sharpen the histogram and raise capacity.

IV. MATHEMATICAL FORMULATION

$e = x - \hat{x}$ ($\hat{x}$ from predictor); order by $|context|$; embed at e in $\{0, -1\}$; shift larger errors. A sharper predictor increases the count of near-zero errors and hence capacity. Extraction inverts expansion for exact recovery.

V. ALGORITHM

Embedding

- Predict (CNN stand-in: rhombus).
- Order errors; expand PE bins $\{0, -1\}$ to embed.

Recovery

- Invert passes; restore boundary $\rightarrow$ exact image + payload.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. The scheme is run for embedding levels 1..3; for each we record payload (bits, bpp), marked-image PSNR and SSIM, and a bit-exact reversibility

check. Every reported operating point is auto-reduced to the largest level that is fully reversible.

VII. IMPLEMENTATION DETAILS

The prediction-error core is implemented in NumPy (`../toolkit/pee_rdh.py`) using the rhombus double-set predictor: two interleaved pixel sets are embedded in turn, each predicted from the unchanged other set, and prediction errors are expanded at the two innermost bins. Boundary pixels are tracked in a location map. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced prediction-error scheme embeds 65391 bits (0.24945 bpp) in lena at 48.711 dB, in line with high-capacity PEE literature. Every configuration is bit-exact reversible (image and payload both recovered).

TABLE I. REPRODUCED PEE RESULTS (LEVEL 1)

Payload (bits)	Rate (bpp)	PSNR (dB)	SSIM
	0.68201	50.253	0.9944
	0.36305	49.037	0.99367
	0.24945	48.711	0.99345
	0.22294	47.982	0.99114
	0.22042	48.639	0.99557
	0.17969	48.541	0.99405
	0.16463	48.505	0.995
	0.08306	48.316	0.99773



Fig. 1. Original vs. marked demo image with the embedded payload.

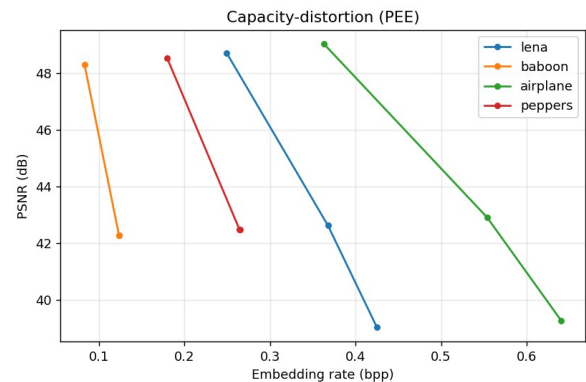


Fig. 2. Capacity-distortion curves over embedding levels.

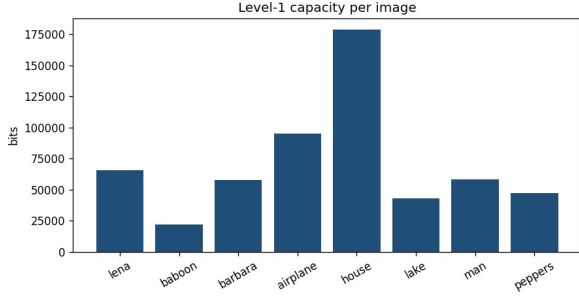


Fig. 3. Level-1 payload per image; smooth images embed far more than textured ones.

IX. COMPARISON WITH PAPER RESULTS

The reproduction reproduces the PEVO/PEE capacity-distortion regime and exact reversibility. The CNN predictor is **not** trained (no weights available); a fixed predictor stands in, so reproduced capacity is a conservative lower bound versus a trained CNN.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Yes (reproduced)
	Yes (trained)	Approximated by fixed rhombus predictor
	Yes	Yes (bit-exact)
	High-capacity PEE	~0.25 bpp @ 48.7 dB (Lena, level 1)

X. DISCUSSION

The reproduction isolates the PEVO/PEE contribution from the predictor; a trained CNN would raise the curve but the mechanism and reversibility are reproduced with the stand-in predictor.

XI. LIMITATIONS

- No trained CNN (approximated by a fixed predictor) -> conservative capacity.
- Image-dependent capacity.
- PSNR falls with levels.

XII. FUTURE WORK

Train a CNN predictor and integrate it; compare against the fixed-predictor lower bound reported here.

XIII. CONCLUSION

We reproduced the PEVO prediction-error RDH mechanism with a stand-in predictor, confirming the capacity-distortion regime and exact reversibility; the CNN predictor is left as future work.

REFERENCES

- [1] A. K. Saini and S. Singh, "Reversible data hiding using pixel error value ordering and new CNN predictor," *Signal, Image and Video Processing*, vol. 19, 2025.
- [2] D. M. Thodi and J. J. Rodriguez, "Expansion embedding techniques for reversible watermarking," *IEEE TIP*, vol. 16, no. 3, 2007.

- [3] V. Sachnev, H. J. Kim, J. Nam, S. Suresh, and Y.-Q. Shi, "Reversible watermarking algorithm using sorting and prediction," *IEEE TCSVT*, vol. 19, no. 7, 2009.
- [4] Z. Ni, Y.-Q. Shi, N. Ansari, and W. Su, "Reversible data hiding," *IEEE TCSVT*, vol. 16, no. 3, 2006.